A counterexample to the R-difference infinite log-concavity conjecture

Independent working note — MathDB 331961

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Task 1: resolve the conjecture. Let

$$R_n = \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k}\frac{1}{2k-1}, \qquad a_n = R_n^2 - R_{n-1}R_{n+1} \quad (n \geq 6).$$

Conjecture 4.8 in the published paper [1] asserts that $\{a_n\}_{n\geq 6}$ is infinitely log-concave. For the log-concavity operator

$$(\mathcal{L}a)_n = a_n^2 - a_{n-1}a_{n+1},$$

we give an exact negative value at the first available center.

Theorem 1. *The published Conjecture 4.8 is false. In fact,*

$$(\mathcal{L}a)_7 = -343413129984 < 0.$$

Both $\{-a_n\}_{n\geq 6}$ and $\{|a_n|\}_{n\geq 6}$ fail the same log-concavity test.

Proof. Direct evaluation of the defining finite binomial sums gives

n	5	6	7	8	9
R_n	329	1359	6001	27759	132689.

Consequently

$$\begin{aligned} a_6 &= 1359^2 - 329 \cdot 6001 = -127448, \\ a_7 &= 6001^2 - 1359 \cdot 27759 = -1712480, \\ a_8 &= 27759^2 - 6001 \cdot 132689 = -25704608. \end{aligned}$$

All three indices are within the stated subsequence. Thus

$$\begin{aligned} a_7^2 - a_6 a_8 &= 1712480^2 - 127448 \cdot 25704608 \\ &= -343413129984 < 0. \end{aligned}$$

The sequence is not even log-concave, so it cannot be infinitely log-concave. Reversing every sign does not change this quadratic expression. Taking absolute values likewise leaves it unchanged because the three displayed terms have the same sign. □

Iteration convention. Even if one tests log-concavity only after first applying $\mathcal{L}$, the assertion fails. Writing $b = \mathcal{L}a$, direct calculation gives

$$b_7 = -343413129984, \quad b_8 = -78797107789056, \quad b_9 = -18331042187598528,$$

and hence

$$b_8^2 - b_7 b_9 = -861363775918512642776924 16 < 0.$$

Only terms of a with indices at least 6 occur. These calculations also use $R_{10} = 649815$ and $R_{11} = 3242377$.

Verification and scope. The standard-library script `verify_counterexample.py` evaluates the binomial sums with rational arithmetic and independently checks them against the source recurrence

$$(n+3)R_{n+3} - (7n+13)R_{n+2} + (7n+15)R_{n+1} - (n+1)R_n = 0,$$

starting from $R_0 = -1, R_1 = 1, R_2 = 7$. The complete exact certificate includes the individual summands. The counterexample matches both the current arXiv source and the published Conjecture 4.8. It does not depend on a negative-term convention, a reversed sign, or an endpoint outside the sequence. The dated source and public-claim audit is saved separately.

References

[1] Brian Yi Sun, *On a ratio monotonicity conjecture of a new kind of numbers*, Journal of Inequalities and Applications **2018**, article 24, Conjecture 4.8. doi:10.1186/s13660-018-1614-1. Earlier version: arXiv:1512.01008.

Task 2: determine the actual log-convexity range. Set $d_n = R_{n-1}R_{n+1} - R_n^2$. Determine the least N for which $\{d_n\}_{n\geq N}$ is positive and strictly log-convex. Give a proof covering every subsequent index.

Task 3: test infinite log-convexity as a replacement. For $(\mathcal{C}d)_n = d_{n-1}d_{n+1} - d_n^2$, determine whether there is one finite starting index such that every iterate $\mathcal{C}^j d$ is positive at all of its remaining indices. If each order instead requires its own starting index, determine how the least such index grows with j.

Task 4: investigate higher Hankel determinants. For $m \geq 2$, determine the signs and eventual nonvanishing of

$$\det\bigl(R_{n+i+j}\bigr)_{0\leq i,j<m}.$$

Obtain effective thresholds in terms of m, and determine whether any fixed shift of the R-sequence has all Hankel minors nonnegative.

Task 5: classify normalizations of the difference. Determine the real parameters α for which the positive sequence $\{n^\alpha |a_n|\}_{n\geq 6}$ is log-concave. Distinguish the exact range for this fixed starting index from the range giving eventual log-concavity.